IGCSE 0607 Practice Questions — E7.5

Non-right-angled triangles

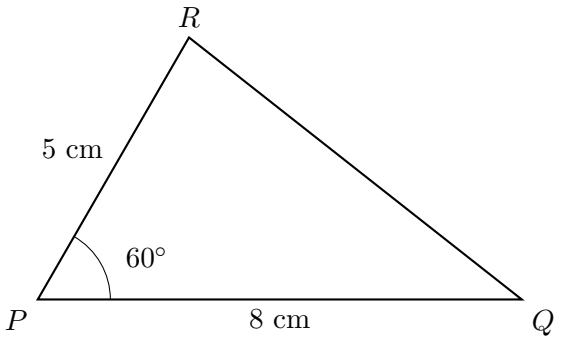
Total Questions: 30 — **Total Marks:** 208 Marks

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- **E7.5** Non-right-angled triangles
  → **Paper 2 (Non-Calculator)** *(2 Qs)* · **Paper 4 (Calculator)** *(28 Qs)*

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1 The diagram shows triangle PQR.
$PQ = 8$ cm, $PR = 5$ cm and angle $QPR = 60°$.



NOT TO SCALE

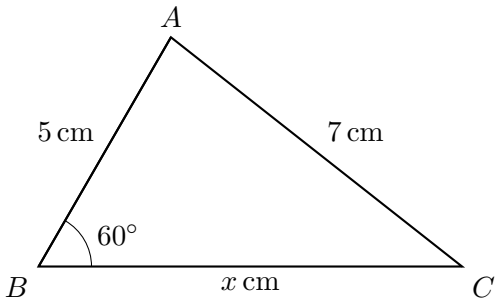
(a) Calculate the exact area of triangle PQR.
Give your answer in the form $a\sqrt{3}$.

..... cm^2 **[2]**

(b) Calculate the exact length of QR.

$QR =$ ..... cm **[3]**

2 In triangle ABC, $AB = 5\,\text{cm}$, $BC = x\,\text{cm}$ and $AC = 7\,\text{cm}$.
Angle $ABC = 60°$.



NOT TO SCALE

(a) Use the cosine rule to show that $x^2 - 5x - 24 = 0$.

[3]

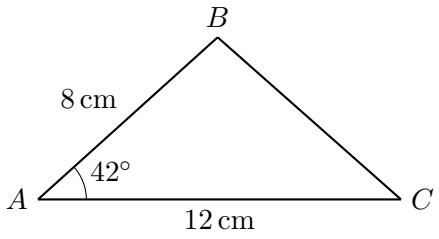
(b) Solve the equation $x^2 - 5x - 24 = 0$ to find the length of BC.

$BC =$ ..... cm **[2]**

(c) Calculate the exact area of triangle ABC.
Give your answer in the form $a\sqrt{b}$, where a and b are integers.

Area = ... cm^2 **[2]**

3 The diagram shows triangle ABC.
$AB = 8\,\text{cm}$, $AC = 12\,\text{cm}$ and angle $BAC = 42°$.



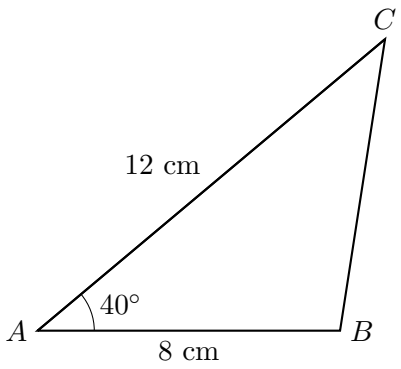
(a) Calculate the area of triangle ABC.

..... cm^2 **[2]**

(b) Calculate the length of BC.

$BC =$ ..... cm **[3]**

4 The diagram shows a triangle ABC.



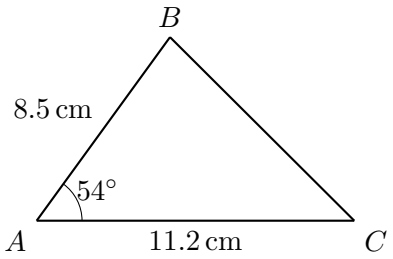
(a) Calculate the area of triangle ABC.

..... cm^2 **[2]**

(b) Calculate the length of BC.

..... cm **[3]**

5 The diagram shows triangle ABC.



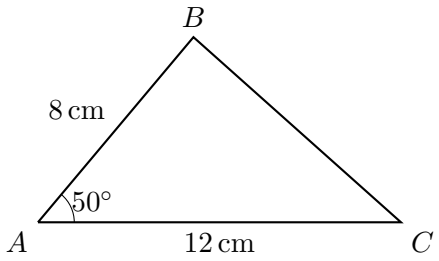
(a) Calculate the area of triangle ABC.

..... cm^2 **[2]**

(b) Calculate the length of BC.

..... cm **[3]**

6 A triangle ABC is shown below.



NOT TO SCALE

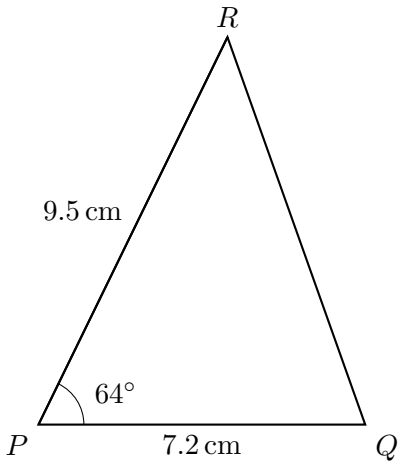
(a) Calculate the area of triangle ABC.

..... cm^2 **[2]**

(b) Calculate the length of BC.

..... cm **[3]**

7 In triangle PQR, $PQ = 7.2\,\text{cm}$, $PR = 9.5\,\text{cm}$ and angle $QPR = 64°$.



NOT TO SCALE

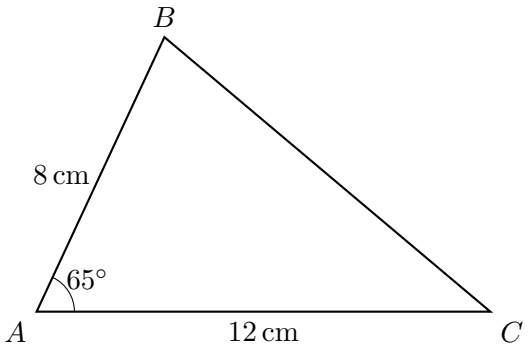
(a) Calculate the area of triangle PQR.

Area = ... cm^2 **[2]**

(b) Calculate the length of QR.

QR = ... cm **[3]**

8 The diagram shows a triangle ABC with $AB = 8\,\text{cm}$, $AC = 12\,\text{cm}$ and angle $BAC = 65°$.



NOT TO SCALE

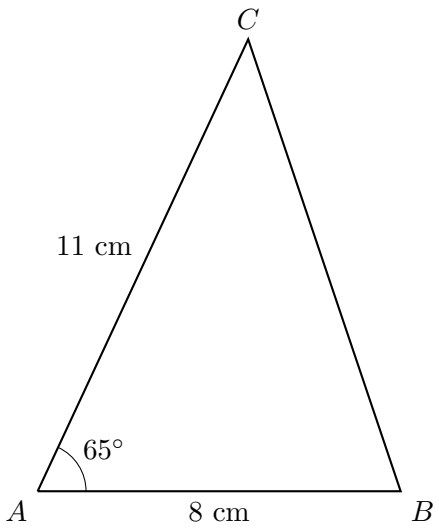
(a) Calculate the length of BC.

$BC =$ ..... cm **[3]**

(b) Calculate the area of triangle ABC.

Area = ... cm^2 **[2]**

9 The diagram shows a triangle ABC.
$AB = 8$ cm, $AC = 11$ cm and angle $BAC = 65°$.



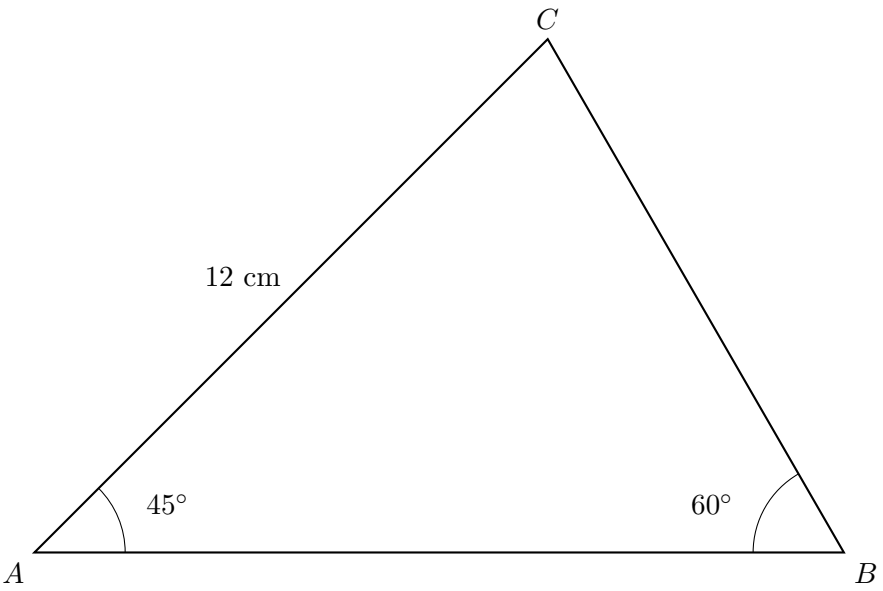
(a) Calculate the area of triangle ABC.

Area = ... cm^2 **[2]**

(b) Calculate the length of BC.

BC = ... cm **[3]**

10 In triangle ABC, $AC = 12$ cm, angle $BAC = 45°$ and angle $ABC = 60°$.



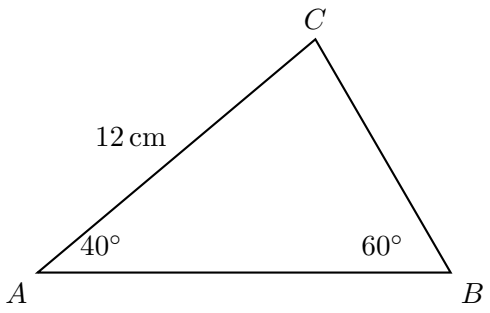
(a) Calculate the length of BC.

..... cm **[3]**

(b) Calculate the area of triangle ABC.

..... cm^2 **[3]**

11 The diagram shows triangle ABC.
$AC = 12$ cm, angle $BAC = 40°$ and angle $ABC = 60°$.



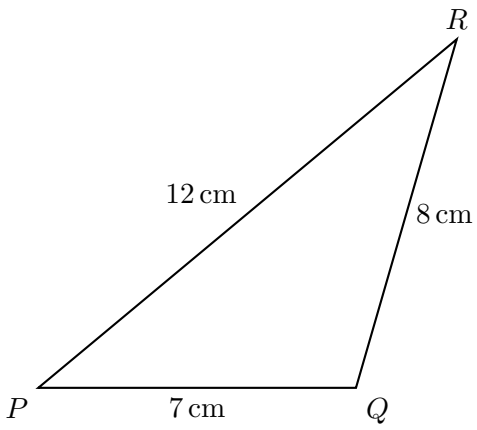
(a) Calculate the length of BC.

$BC =$ ..... cm **[3]**

(b) Calculate the area of triangle ABC.

Area = ... cm^2 **[3]**

12 The diagram shows triangle PQR.
$PQ = 7$ cm, $QR = 8$ cm and $PR = 12$ cm.



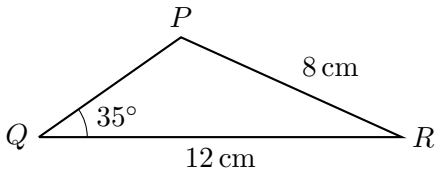
(a) Calculate the largest angle in triangle PQR.

..... ° **[3]**

(b) Calculate the area of triangle PQR.

Area = ... cm^2 **[2]**

13 In triangle PQR, $PR = 8$ cm, $QR = 12$ cm and angle $PQR = 35°$.
Angle QPR is an obtuse angle.



NOT TO SCALE

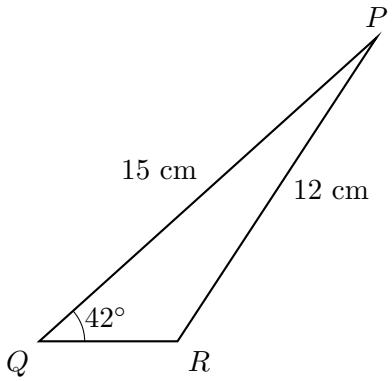
(a) Calculate the size of angle QPR.

..... ° **[4]**

(b) Calculate the area of triangle PQR.

..... cm^2 **[2]**

14 The diagram shows a triangle PQR.
The length of PQ is 15 cm and the length of PR is 12 cm.
Angle $PQR = 42°$.
Angle PRQ is obtuse.



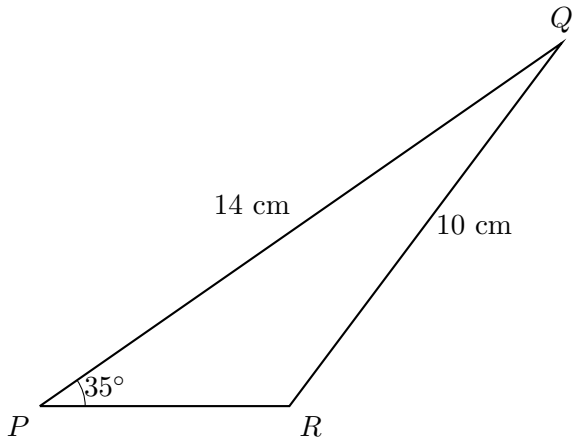
(a) Calculate angle PRQ.

..... ° **[3]**

(b) Calculate the length of QR.

..... cm **[3]**

15 The diagram shows triangle PQR.
$PQ = 14$ cm, $QR = 10$ cm and angle $QPR = 35°$.
Angle PRQ is obtuse.



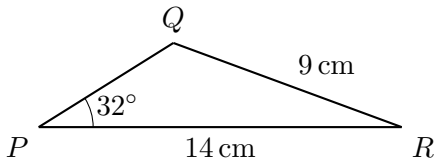
(a) Calculate angle PRQ.

Angle $PRQ =$ ..... ° **[3]**

(b) Calculate the area of triangle PQR.

Area = ... cm^2 **[3]**

16 The diagram shows triangle PQR.
The angle PQR is obtuse.



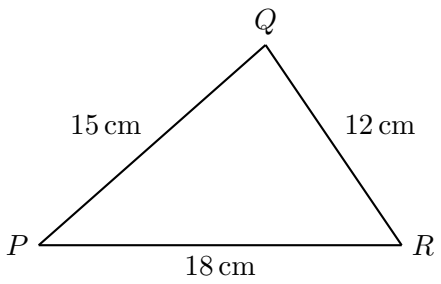
(a) Calculate the obtuse angle PQR.

..... ° **[4]**

(b) Calculate the area of triangle PQR.

..... cm^2 **[3]**

17 The diagram shows triangle PQR.
$PQ = 15\,\text{cm}$, $QR = 12\,\text{cm}$ and $PR = 18\,\text{cm}$.



(a) Use the cosine rule to show that angle $PQR = 82.8°$, correct to 1 decimal place.

[3]

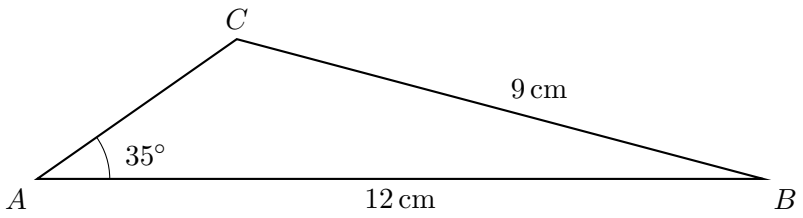
(b) Calculate the area of triangle PQR.

..... cm^2 **[2]**

(c) Calculate the shortest distance from R to the line PQ.

..... cm **[2]**

18 In triangle ABC, $AB = 12\,\text{cm}$, $BC = 9\,\text{cm}$ and angle $BAC = 35°$.



NOT TO SCALE

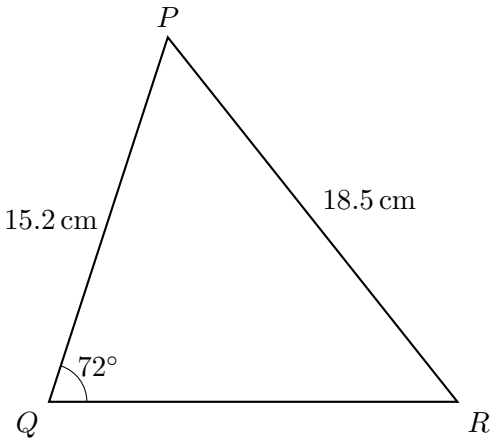
(a) Calculate the two possible values of angle ACB.
Give your answers correct to 1 decimal place.

..... ° **[4]**

(b) Given that angle ACB is obtuse, calculate the area of triangle ABC.

Area = ... cm^2 **[3]**

19 The diagram shows a triangle PQR.
$PQ = 15.2\,\text{cm}$, $PR = 18.5\,\text{cm}$ and angle $PQR = 72°$.



NOT TO SCALE

(a) Calculate the size of angle PRQ.

Angle $PRQ =$ ..... ° **[3]**

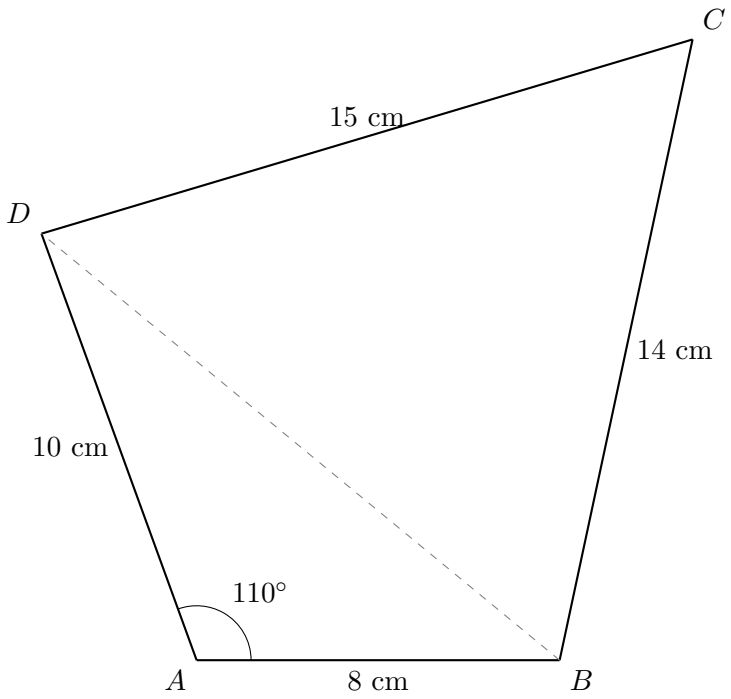
(b) Calculate the length of QR.

$QR =$ ..... cm **[3]**

(c) Calculate the shortest distance from P to the line QR.

Distance = ... cm **[2]**

20 The diagram shows a quadrilateral $ABCD$.
$AB = 8$ cm, $AD = 10$ cm, $BC = 14$ cm and $CD = 15$ cm.
Angle $BAD = 110°$.



(a) Calculate the length of the diagonal BD.

..... cm **[3]**

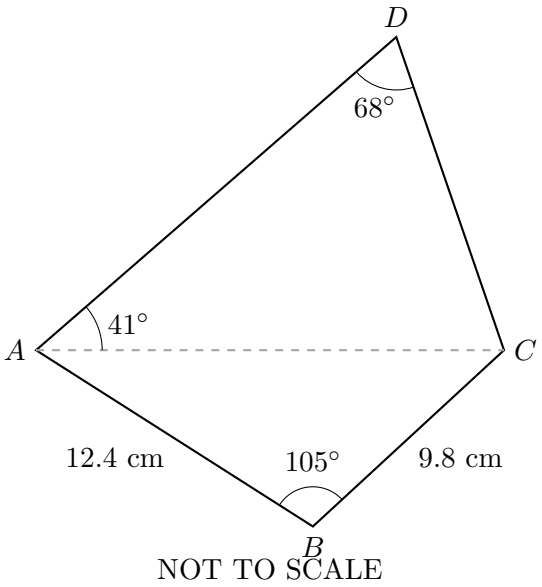
(b) Calculate the angle BCD.

..... ° **[3]**

(c) Calculate the area of the quadrilateral $ABCD$.

..... cm^2 **[3]**

21 The diagram shows a quadrilateral $ABCD$.
$AB = 12.4$ cm, $BC = 9.8$ cm and angle $ABC = 105°$.
Angle $CAD = 41°$ and angle $ADC = 68°$.



NOT TO SCALE

(a) Calculate the length of AC.
Give your answer correct to 3 significant figures.

$AC =$ ..... cm **[3]**

(b) Calculate the length of CD.
Give your answer correct to 3 significant figures.

$CD =$ ..... cm **[3]**

(c) Calculate the total area of the quadrilateral $ABCD$.

..... cm^2 **[3]**

22 In triangle ABC, $AB = 15$ cm, $BC = 12$ cm and angle $BAC = 48°$.

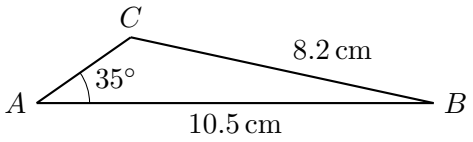
(a) Calculate the two possible values of angle BCA.

and ..... **[3]**

(b) Hence, calculate the difference between the two possible areas of triangle ABC.

..... cm^2 **[4]**

23 In triangle ABC, $AB = 10.5\,\text{cm}$, $BC = 8.2\,\text{cm}$ and angle $BAC = 35°$.
Angle BCA is obtuse.



(a) Calculate the size of angle BCA.

..... ° **[3]**

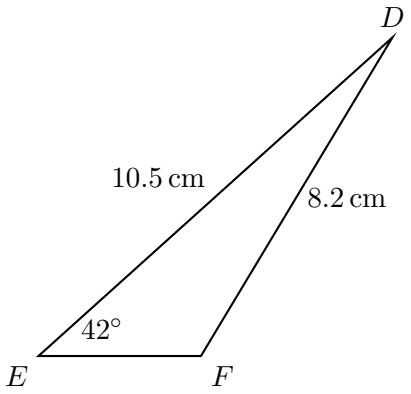
(b) Calculate the length of AC.

$AC =$ ..... cm **[3]**

(c) A point D lies outside the triangle ABC such that $CD = 6.4\,\text{cm}$ and angle ACD is acute.
The area of triangle ACD is $8.5\,\text{cm}^2$.
Calculate the size of angle ACD.

..... ° **[2]**

24 The diagram shows triangle DEF.
$DE = 10.5$ cm, $DF = 8.2$ cm and angle $DEF = 42°$.
Angle DFE is an obtuse angle.



(a) Calculate angle DFE.

Angle $DFE =$ ..... ° **[3]**

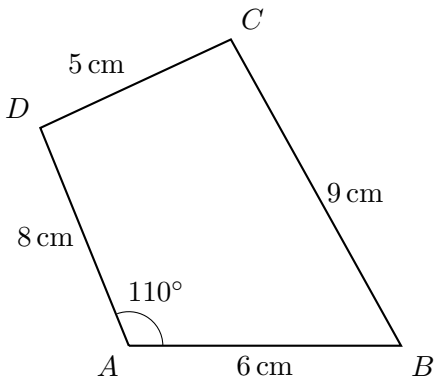
(b) Calculate the length of EF.

$EF =$ ..... cm **[3]**

(c) Calculate the shortest distance from D to the line EF.

..... cm **[2]**

25 The diagram shows a quadrilateral $ABCD$.
$AB = 6$ cm, $AD = 8$ cm, $DC = 5$ cm, and $BC = 9$ cm.
Angle $DAB = 110°$.



NOT TO SCALE

(a) Calculate the length of the diagonal BD.

..... *cm* **[3]**

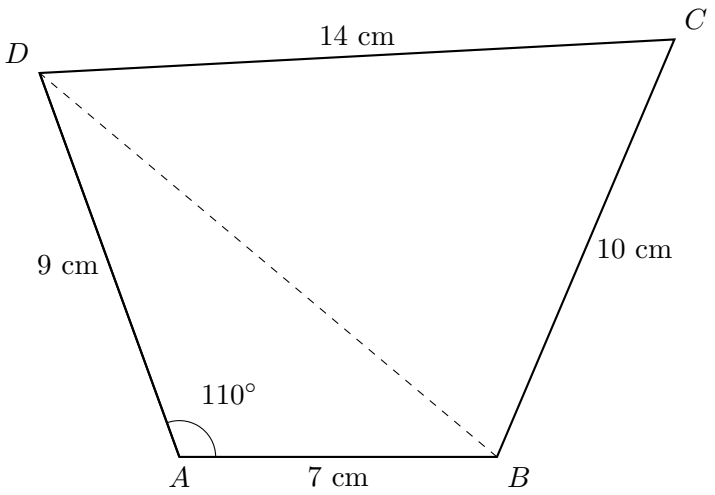
(b) Calculate the size of angle BCD.

..... ° **[3]**

(c) Calculate the total area of the quadrilateral $ABCD$.

..... cm^2 **[3]**

26 The diagram shows a quadrilateral $ABCD$.
$AB = 7$ cm, $AD = 9$ cm, $BC = 10$ cm and $CD = 14$ cm.
Angle $BAD = 110°$.



(a) Calculate the length of BD.

..... cm **[3]**

(b) Calculate angle BCD.

..... ° **[3]**

(c) Calculate the total area of the quadrilateral $ABCD$.

..... cm^2 **[3]**

27 The diagram shows a quadrilateral $ABCD$.
$AB = 7.4$ cm, $BC = 9.1$ cm, and $CD = 10.5$ cm.
Angle $ABC = 112°$ and angle $CAD = 43°$.

(a) Calculate the length of AC.

..... cm **[3]**

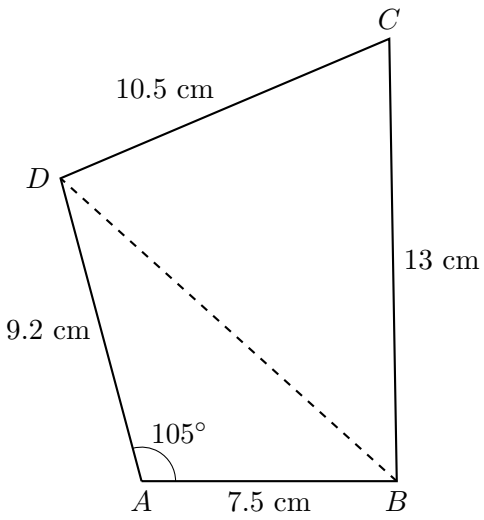
(b) Calculate the acute angle ADC.

..... ° **[3]**

(c) Calculate the shortest distance from D to the line AC.

..... cm **[3]**

28 The diagram shows a quadrilateral $ABCD$.
$AB = 7.5$ cm, $AD = 9.2$ cm and angle $BAD = 105°$.
$BC = 13$ cm and $CD = 10.5$ cm.



(a) Calculate the length of BD.

$BD =$ ..... cm **[3]**

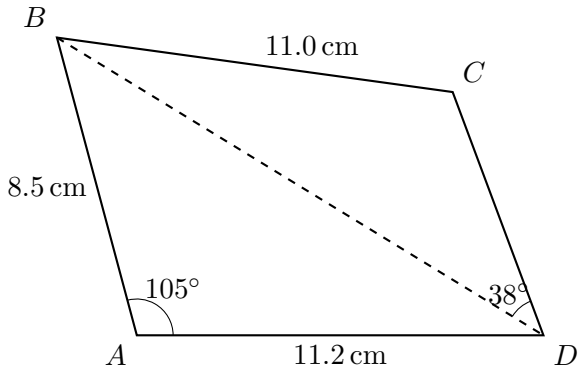
(b) Calculate angle BCD.

Angle $BCD =$ ..... ° **[3]**

(c) Calculate the area of the quadrilateral $ABCD$.

Area = ... cm^2 **[3]**

29 The diagram shows a quadrilateral $ABCD$.
$AB = 8.5\,\text{cm}$, $AD = 11.2\,\text{cm}$ and angle $BAD = 105°$.
$BC = 11.0\,\text{cm}$ and angle $BDC = 38°$.
Angle BCD is obtuse.



NOT TO SCALE

(a) Calculate the length of BD.

$BD =$ ..... cm **[3]**

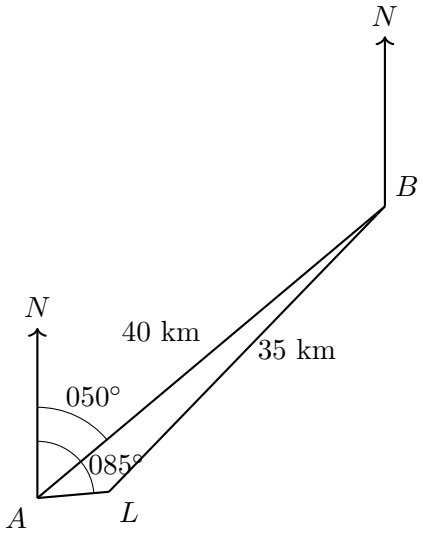
(b) Calculate the size of angle BCD.

Angle $BCD =$ ..... ° **[3]**

(c) Calculate the area of the quadrilateral $ABCD$.

Area = ... cm^2 **[4]**

30 Ship A and Ship B are at sea.
Ship B is 40 km from Ship A on a bearing of 050°.
A lighthouse L is 35 km from Ship B.
The bearing of L from A is 085°.
Given that the angle ALB is obtuse:



(a) Calculate the obtuse angle ALB.

..... ° **[4]**

(b) Calculate the distance AL.

..... km **[3]**

(c) Calculate the bearing of L from B.

..... ° **[3]**